

GUIDANCE ON THE USE OF SIGNIFICANT DIGITS IN THE TESTING LABORATORY

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TABLE OF CONTENTS

1.0	PURPOSE	3
2.0	SCOPE	3
3.0	DEFINITIONS.....	3
4.0	GENERAL	4
4.1	Significant Digits:	4
5.0	GUIDANCE ON THE USE OF SIGNIFICANT DIGITS IN THE RECORDING OF RESULTS OF MEASUREMENTS AND CALCULATIONS.....	5
5.1	Data Recording - General:.....	5
5.2	Analog Display Measurements:.....	6
5.3	Digital Display Measurements:.....	6
5.4	Adjustment of Significant Digits Recorded:	6
6.0	GUIDANCE ON DETERMINING CONFORMANCE WITH SPECIFICATIONS	7
6.1	General:	7
6.2	Test Parameters:	7
6.3	Tolerances	9
6.4	Conformance Criteria:	10
7.0	GUIDANCE ON TIME MEASUREMENTS.....	11
7.1	Time Measurements:	11
8.0	CALCULATIONS AND SIGNIFICANT DIGITS	12
9.0	ROUNDING:	14
	REFERENCES:	15

1.0 PURPOSE

- 1.1.1 The purpose of this document is to provide guidance on the determination of the number of significant digits from specifications and use of significant digits when:
- A. Selecting test measurement equipment,
 - B. Establishing test parameters and parameter tolerances,
 - C. Recording data,
 - D. Performing calculations and
 - E. Determining conformance of that data or results of calculations with specifications.

2.0 SCOPE

- 2.1.1 The scope of this guidance covers the following situations.
- A. Indications of precision and accuracy are evident from the statement of requirements in the specification through use of scientific or standard notation.
 - B. Interpreting test parameters and compliance criteria given in specifications when no clear indication of expected precision and accuracy is expressed.
 - C. Interpreting test parameters and compliance criteria given in standards or specifications when references are to trade sizes.
 - D. Comparison of the data employing significant digits to the specifications.
- 2.1.2 Once a determination of the significant digit requirements applicable to the specifications has been made, the applicable mathematical rules are provided in this document.

3.0 DEFINITIONS

- 3.1.1 General – Several of the terms noted below have different definitions when used in the documents noted in the “Reference Section” of this document. Care should be taken in use of these terms and to ensure that the correct definition is used and understood in written communications or discussions.

Accuracy – Closeness of agreement between a measured quantity value and a true quantity value of a measurand.

Note 1: The concept “measurement accuracy” is not a quantity and is not given a numerical quantity value. A measurement is said to be more accurate when it offers a smaller measurement error.

Note 2: The “measurement accuracy” should not be used for “measurement trueness” and the term “measurement precision” should not be used for “measurement accuracy”, which, however, is related to both these concepts.

Note 3: “Measurement accuracy” is sometimes understood as closeness of agreement between measured quantity values that are being attributed to the measurand.

Measurand – Quantity intended to be measured.

Precision – The closeness of agreement between indications or measured quantity values obtained by replicate measurements on the same or similar objects under specified conditions.

Note 1: Measurement precision is usually expressed numerically measure of

imprecision, such as standard deviation, variation, or coefficient of variation under the specified conditions of measurements.

Note 2: The “specified conditions” can be, for example, repeatability conditions of measurement, intermediate precision conditions of measurement, or reproducibility conditions of measurement.

Note 3: Measurement precision is used to define measurement repeatability, intermediate measurement precision, and measurement reproducibility.

Note 4: Sometimes “measurement precision” is erroneously used to mean “measurement accuracy”.

Resolution –Smallest change in a quantity being measured that causes a perceptible change in the corresponding indication.

Note 1: Resolution can depend on, for example, noise (internal or external) or friction. It may also depend on the value of a quantity being measured.

Note 2” The number of digits displayed on a digital readout is not necessarily an indication of the resolution of the measurement.

Specifications – Testing requirements contained in a standard or a customer’s testing specification.

Testing Laboratory (TL) – A local testing laboratory or location where testing or testing related processes occur.

Tolerance – The amount a measured value can vary and still be considered as meeting the specification. (IEEE SI 10-2002: The amount by which the value of a quantity is allowed to vary, thus the tolerance is the algebraic difference between the maximum and minimum limits.)

Absolute Method - Specification limits of a value where a maximum or minimum designation represent an absolute limit of exactly the specified value and for purposes of determining conformance with specifications, an observed value or a calculated value is to be compared directly with the specified limit.

Rounding Method - Specification limits of a value where a maximum or minimum designation are taken to imply that, for the purposes of determining conformance with specifications, an observed value or a calculated value should be rounded to the number of significant digits of the specification limit, and then compared with the specification limit.

Consensus Limits – A consensus of judgement of what the limits of the test result should be. Consensus limits are typically established as the result of a committee action and typically employee a safety or design factor.

4.0 GENERAL

4.1 Significant Digits:

- 4.1.1 **Scientific Notation** – Scientific notation is a way to format numbers that clearly expresses the magnitude and precision of a measured value. Scientific notation consists of a positive or negative real number with a magnitude greater than 1 and less than 10 in combination with a positive or negative integer exponent of the power of ten. The number of digits used in the real number signifies the precision of the value and the exponent indicates the magnitude. The exponent part of the number can be written either as the integer 10 with a superscript exponent or E-format. Examples are:

$$2.54 \times 10^{+2} = 2.54 \text{ E } + 2$$

$$1.492 \times 10^{-3} = 1.492 \text{ E } - 3$$

4.1.2 **Standard Notation** – Standard notation is a number whose digits and placement of the decimal point convey the magnitude and precision of measured value, for example 2000.3, 300.04 and 0.045. In using standard notation, care must be taken to prevent ambiguity about the precision of the number. The following rules apply:

- Non-zero digits are always significant
- Zeroes with non-zero digits on both sides are significant
- Leading zeroes are not significant.
- Trailing zeroes are significant if to the right or left of the decimal.
- Trailing zeroes where there is no decimal are not significant.

Examples:

- 2000 has one significant digit
- 2000. has four significant digits (note the decimal point following the final 0).
- 2000.00 has six significant digits.

Note - At times it is difficult to determine if standard notation was used or not used. In such cases it is necessary to use knowledge of the test procedure or other information to determine the intent.

Examples:

- 321.56 has an accuracy of 5 significant digits and a precision of 2 decimal places.
- 3.2156 has an accuracy of 5 significant digits and a precision of 4 decimal places.
- 321560 has an accuracy of 5 significant digits and a precision of tens (assuming the rightmost zero is not significant).
- 0.000003 has an accuracy of 1 significant digit and a precision of 6 decimal places. Based on stated units, 0.000003 in. is precise to the nearest 10^{-6} in = 3 μ in is precise to the nearest μ in.
- 325,000,000 has an accuracy of 3 significant digits and a precision to the nearest million.

4.1.3 **Integer Numbers** – Integer numbers are exact numbers obtained by counting or by definition, such as for some conversion factors. All digits are significant. Determining if a number is an integer or not depends upon knowledge of what the number represents.

Examples:

- 6000 cycles has four significant digits
- 10 samples has 2 significant digits

Note - Scientific notation is the number format that most clearly indicates the number of significant digits along with the magnitude of the number. Use of standard notation can lead to ambiguity about what digits are significant unless care is taken in how the number is written.

5.0 GUIDANCE ON THE USE OF SIGNIFICANT DIGITS IN THE RECORDING OF RESULTS OF MEASUREMENTS AND CALCULATIONS

5.1 Data Recording - General:

5.1.1 Correct recording of test results, requires knowledge of the capability of measuring instruments. Correct recording of results of calculations requires that rules for

calculations involving significant digits be followed.

- 5.1.2 The results of measurements and calculations should be recorded with the number of significant digits that reflect the precision and accuracy of the result so that correct conclusions can be drawn from the value presented. Recording too many digits implies false precision. Recording of too few digits results in round-off errors.
- 5.1.3 The number of significant digits used to report results of measurements and calculations may be reduced to reflect the precision and accuracy contained in a specification governing the testing and interpretation of the results.
- 5.1.4 The number of digits that are significant are the number of digits needed to express the number within the uncertainty of the measurement or calculation.

5.2 Analog Display Measurements:

- 5.2.1 When recording observed values, the number of digits to be recorded are all exactly known digits plus one additional digit that can be reasonably estimated, otherwise the reading is recorded to the nearest graduation. Refer to 5.4 on adjustment of significant digits recorded for exceptions.

Example – An observed reading from an analog voltmeter that has graduations to the tenth of a volt was recorded as 8.37 volts because the observed reading was between the 8.3 and 8.4 volts graduation and it was possible to estimate seven tenths of the way between the graduations.

Example – A measurement was taken using a steel rule with 1 mm graduations. An observed reading was recorded as 12.2 cm because the observed reading was between 12.2 and 12.3 graduations, but closer to 12.2. It was not possible to reasonably estimate the distance between 12.2 and 12.3 graduations, so the measurement was report to the closest graduation

5.3 Digital Display Measurements:

- 5.3.1 When recording observed values, the number of significant digits to be recorded are all stabilized digits displayed. Refer to 5.4 on adjustment of significant digits recorded for exceptions.

Example - An observed reading from a digital voltmeter that can display to the one hundredth of a volt has stabilized digits for all but the one hundredth volt digit, which continuously changed. A reading of 19.7 volts was recorded.

5.4 Adjustment of Significant Digits Recorded:

- 5.4.1 Well designed and manufactured measurement instruments should provide a display of results commensurate with the instrument accuracy. However, this is not always the case and is especially true of measurement systems that use a universal display in conjunction with separate and/or interchangeable measurement modules. Examples of such systems are data loggers with interchangeable measurement cards and computerized data acquisition systems that gather information from a diversity of measurement instruments. For such measurement instruments, it is necessary to determine what the precision and accuracy specifications are for the measurement module and record the observed reading to the number of significant digits that are meaningful.

Note – Often, general purpose data acquisition software has provision for adjusting the display format to reflect the accuracy and precision of the measured quantity. However, programmers of the systems do not always make use of this feature.

Example – A digital pressure transducer is connected to a data logger that has a 12 digit display. Data is transmitted from the pressure transducer to the data logger over a

digital bus commensurate with the 12 digit display. The pressure transducer operates over a range of 0 to 100 kPa with an accuracy of ± 0.1 kPa. How many significant digits would be meaningful in a displayed value? The displayed value would be meaningful to one significant digit to the right of the decimal point, for example 81.3 kPa. Displayed values containing more digits to the right of the decimal point should be rounded to one digit to the right of the decimal point. If the data logger has a display formatting function, it would be convenient to program the display to do the rounding and, then, display the value to no more than one decimal place.

Example – An analog pressure transducer is connected to a data logger that has a 12 digit display. The output of the pressure transducer is a 0-5.00 volt signal that represents 0-100 kPa with an accuracy of ± 1 kPa. The output of the pressure transducer is feed into a voltage measurement card mounted in a slot in the data logger. Within the data logger data is transmitted over a digital bus commensurate with the 12 digit display. The accuracy of the voltage measurement card is $\pm 1\%$ of full scale. The voltage card is set to the 0-10.0 volt range. The laboratory metrologist calculates the uncertainty of measurement associated with the measuring instruments used and finds that the uncertainty is ± 2 kPa, $k = 2$, 95% confidence interval. How many significant digits would be meaningful in a displayed value? The displayed value would be significant to one's digit. Therefore, any displayed value in the range of 1 to 100 kPa would be significant, for example 100 kPa (three significant digits), 98 kPa (two significant digits) and 7 kPa (one significant digit).

6.0 GUIDANCE ON DETERMINING CONFORMANCE WITH SPECIFICATIONS

6.1 General:

- 6.1.1 When standards and specifications are written, it would be useful if the test parameters and compliance criteria contained in the standards and specifications were written to indicate the precision and accuracy associated with the criteria. While some standards and specifications explicitly give tolerances on numerical values, or imply expected precision and accuracy through the number of significant digits given, many standards give no guidance whatsoever. Still other standards and specifications give numerical values with the expectation of the value being absolute.
- 6.1.2 Section 6.0 of this document is intended to give guidance on interpreting test parameters and compliance criteria given in standards and specifications when no clear indication of expected precision and accuracy is expressed.

6.2 Test Parameters:

- 6.2.1 Test parameters are specifications for various conditions that affect the outcome of a test. Typical test parameters that may be specified are:
 - A. Temperature
 - B. Humidity
 - C. Barometric pressure
 - D. Power source voltage
 - E. Power source frequency
 - F. Power source total harmonic distortion
 - G. Air velocity
 - H. Force applied
 - I. Test time – duration
 - J. Energy content of fuel gas

K. Dimensions of test fixtures

L. Current

M. Power factor

N. Weight

6.2.2 In many instances, test parameters are provided with tolerances within which the value may vary. For example 25 ± 5 °C would have a set point of 25 °C and permit the temperature to vary between 20 and 30 °C. It is not intended that the set temperature be at the extremes of the tolerance range.

6.2.3 When explicit indication of what is meant by the specification is not provided, having background understanding for the development of the standard and specification is invaluable with understanding what was intended. Even without explicit indication, it may be possible to surmise the intent from the context in which the test parameter is written.

Note: The determination of the intent should be used with caution as data showing the sensitivity of the test result to variations in the test parameter may not be available to support the implied tolerances.

6.2.4 **Single point specification** - There are two predominant views on how single point specifications should be treated. There is the **absolute method** whereby single point number indicates the exact value. There is the **rounding method** whereby the number of significant digits used in the specification may be interpreted to reflect the intended precision and accuracy.

6.2.5 Following the **absolute method**, a single point specification is treated as though it had an infinite number of significant digits.

6.2.6 Following the **rounding method**, the implied precision of a single point test specification is the number of digits in the specification $\pm$ half the least significant digit. In some situations, this method may be interpreted as defining a default tolerance in the absence of any other information.

Example 1

Absolute Method - The specification that a product be tested while connected to a 120 volt power source has the expectation that the voltmeter used to measure the power source voltage will indicate 120 volts, 120.0 volts, 120.00 volts, etc. within the capability of the measuring instrument used.

Rounding Method - The specification that a product be tested while connected to a 120 volts power source has the expectation that the voltmeter used to measure the power source voltage will indicate 120 volts when rounded to the nearest volt and may drift ± 0.5 volts (i.e. between 119.5 volts and 120.5 volts), assuming that the 0 is significant.

Example 2

Absolute Method - The specification that a product be tested while connected to a 230 volts power source has the expectation that the voltmeter used to measure the power source voltage will indicate 230 volts, 230.0 volts, 230.00 volts, etc. within the capability of the measuring instrument used.

Rounding Method - The specification that a product be tested while connected to a 230 volt power source has the expectation that the voltmeter used to measure the power source voltage will indicate 230 volts when rounded to the nearest volt and may drift ± 0.5 volts (i.e. between 229.5 volts and 230.5 volts), assuming that the 0 is significant.

Example 3

Absolute Method - The specification that a product tested while connected to 50 Hz power source has the expectation that the frequency meter indicate 50 Hz, 50.0 Hz,

50.00 Hz, etc. within the capability of the measuring instrument used.

Rounding Method – The specification that a product tested while connected to 50 Hz power source has the expectation that the frequency meter indicate 50 Hz when rounded to the nearest Hertz and may drift ± 0.5 Hz (i.e. between 49.5 Hz and 50.5 Hz), assuming that the 0 is significant.

- 6.2.7 For both methods, in absence of an accuracy specification, the accuracy of the measurement equipment used would default to the agreed upon accuracy for making the measurement. An example is IECCEB Scheme CTL DSH-251B.
- 6.2.8 **Limit Specifications** – There are also two predominant views on how limit specifications should be treated. There is the **absolute method** whereby the limit number indicates the exact value. There is the **rounding method** whereby the number of significant digits used in the limit specification may be interpreted to reflect the intended precision and accuracy.
- 6.2.9 Following the **absolute method**, a limit specification is treated as though it had an infinite number of significant digits.
- 6.2.10 Following the **rounding method**, the implied precision of a limit specification is the number of digits in the specification.

Example 1

Absolute Method – The specification that a product be tested with an ambient humidity between 40 % RH and 60% RH has the expectation that the humidity meter indicate not less than 40 % RH and not more than 60% RH, as displayed on the measuring instrument. Humidity measurements of 39.9 % RH, 60.1 % RH, 39.99 % RH, 60.01 % RH and so on would be considered outside the limits.

Rounding Method – The specification that a product be tested with an ambient humidity between 40 % RH and 60% RH has the expectation that the humidity meter indicate not less than 40 % RH and not more than 60% RH when rounded to the nearest 1% RH. Instrument readings of 39.6 % RH and 60.4 % RH would be rounded to 40 % RH and 60 % RH, respectively, and would be considered within specification.

- 6.2.11 In absence of an accuracy specification, the accuracy of the measurement equipment used would default to the agreed upon accuracy for making the measurement. An example is IECCEB Scheme CTL DSH-251B..

6.3 Tolerances

- 6.3.1 In some cases, tolerances of test specifications are explicitly stated. See 6.2.2
- 6.3.2 In the remainder of the cases, a single point test specification is provided without tolerances.
 - A. It may be possible to surmise a tolerance from the context in which the test parameter is written.
 - B. In some situations, the rounding method may be interpreted as defining a default tolerance in the absence of any other information because of the implied precision, which is the number of digits in the specification $\pm$ half the least significant digit.
- 6.3.3 Tolerance around a specification set point indicates that the test value is to be set at the specified value, but may vary from it during the course of the testing due to control or environmental fluctuations, or other causes. It is not intended to establish a range of set points up to the extremes of the tolerance range to facilitate compliance with test results.
- 6.3.4 When attempting to assign a tolerance where none is specified considerations include:
 - A. Data showing the sensitivity of the test result to variations in the test parameter may not be available to support the implied or assigned tolerances.
 - B. Tolerances may not be equal around the test parameter, i.e. 10 ± 0.5 mm where

a value must err toward ensuring determination of a compliant result vs. a false noncompliance.

- C. Tolerances for test fixture specifications need to consider manufacturing tolerances, the cost to manufacture to those tolerances, the materials used to construct the test fixture, environmental impact on those materials (expansion/contraction due to temperature and/or humidity), etc.
- D. Recognition of the test measurement equipment resolution used in making the measurement. This would include consideration of the resolution of test equipment commonly available at the time the specification was published.

Example - An analog volt meter for a dielectric withstand tester may have a resolution of 50 V on the 1500 V scale and represents equipment commonly available when the specification was published. Today, a digital display for a similar tester may have a display resolution of 1 V. Both devices may be used for the test if the standard does not identify measurement resolution.

- 6.3.5 Additional Safety considerations exist as follows: Staff should understand the effect that tolerance variance could have on the safe outcomes of a test procedure.

6.4 Conformance Criteria:

- 6.4.1 Conformance criteria are specifications that are used to make pass/fail compliance decisions. The results of tests are compared to conformance criteria to determine if the product meets the test specification.
- 6.4.2 In absence of an accuracy specification for measurement results, the accuracy of the measurement equipment used would default to the agreed upon accuracy for making the measurements. An example is IEC CB Scheme CTL DSH-251B.
- 6.4.3 In applying acceptance criteria when the precision and accuracy required is not explicitly stated, additional consideration is required of whether the limit is an **absolute limit or a consensus limit**.
- 6.4.4 With an **absolute limit**, an observed test result or calculated value is not to be rounded, but is to be compared directly with the specified limit value. Conformance or non-conformance with the limit is based on this comparison.
- 6.4.5 Absolute limits are often based on statistical data that assures with a defined confidence level that something will or will not happen, and does not include a design or safety factor.

Example – The leakage current measured shall not exceed 0.5 mA. In this case any test result > 0.5 mA is considered a nonconforming result. Nonconforming results include: 0.51 mA, 0.501 mA, 0.5001 mA,

- 6.4.6 **Consensus limits** are based on a consensus of judgement of what the limits of the test result should be. Consensus limits are typically established as the result of a committee action and typically employ a safety or design factor. Exceeding the limit by a small amount does not result in an imminent hazard.
- 6.4.7 When comparing the results of tests and calculations to consensus limits, the result is rounded to the same number of significant digits contained in the specification. Conformance or non-conformance with the limit is based on this comparison.

Example – The temperature on the coil shall not exceed 105°C. A measured temperature of 105.4°C rounds to 105°C and, therefore, is considered a conforming. Whereas a measured temperature of 105.6°C is considered a nonconformance, as it rounds to 106°C.

Example – The insulation shall withstand a test voltage of twice rated voltage plus 1000 volts. The rated voltage is 230 volts. Therefore, the required test voltage is 1460 volts. A product withstanding 1459.5 volts as indicated on the meter is considered pass as it

rounds to 1460 volts. A product withstanding 1459.49 volts as indicated on the meter is considered a fail as the voltage rounds to 1459 volts.

7.0 GUIDANCE ON TIME MEASUREMENTS

7.1 Time Measurements:

7.1.1 Questions have been made on how to interpret certain time specifications found in standards and what type of instrumentation should be used to make the measurements. The way the time interval specifications are written in many standards is inadequate. Many time intervals were defined without regard to uncertainty of measurement or significant digits. The intervals were defined based on what might be called a “common notion” of time.

Examples:

- A. High Voltage Withstand or Dielectric test – Typically, a voltage is gradually applied. The product must withstand the voltage for one minute. If breakdown happens in anything less than 60 seconds, it is considered a nonconformance. Breakdown at 61 seconds is acceptable. One needs a calibrated stopwatch accurate to the nearest second as a difference of one second is critical.
- B. Sample conditioning - Must be conditioned for 48 hours. The expectation is that the samples are placed in the specified environment and will be ready two days later at or after the same time of day the conditioning started. Calendars in conjunction with wall clocks of commonly available capability are adequate.
- C. 72 hour motor locked rotor test - The expectation is that the test is run for a minimum of 72 hours as measured to the minute. A calibrated timer accurate to the nearest minute is needed.
- D. 18 day motor locked rotor test - The expectation is that the test will be finished 18 days later at or after the same time of day the test was started. Calendars in conjunction with wall clocks of commonly available capability are adequate.

8.0 CALCULATIONS AND SIGNIFICANT DIGITS

- 8.1.1 **General** – Following are the common rules for calculations involving significant digits. Additional rules for trigonometric functions, logarithms, and exponentials are available but have not been included here for brevity.
- 8.1.2 In intermediate calculations, retain at least one extra digit than is specified for the rules of the mathematical operation to avoid round-off errors that can accumulate to produce inaccurate results. Round off the end value of the calculation to the correct number of significant digits.
- 8.1.3 **Addition and Subtraction** – When adding or subtracting numbers, the answer should not have more decimal places than the number with the least decimal places. When doing the calculation keep all decimal places and round the result.

Examples:

$$12.25 + 8.4 + 7.23 = 27.9$$

$$(3.45 \text{ E } + 5) + (4.1 \text{ E } + 4) = 3.9 \text{ E } + 5$$

- 8.1.4 **Multiplication and Division** – When multiplying or dividing numbers, the answer should have not more than the same number of significant digits as the number with the least number of significant digits.

Examples:

$$3.293 \times 3.5 = 12 \quad \text{four significant digits} \times 2 \text{ significant digits} = 2 \text{ significant digits}$$

- 8.1.5 **Multiplication and Division By An Integer Or Constant** – When multiplying a number by an integer or conversion constant, the answer should have the same number of significant digits as the number. Numbers that are exact are treated as though they consist of an infinite number of additional significant digits. When a count (an integer) is used in a computation with a measurement, the number of significant digits in the answer is the same as the number of significant digits in the measurement.

Examples:

$$1345.87 \text{ in.} \div 12 \text{ in/ft} = 112.156 \text{ ft.}$$

$$\text{AVERAGE} = (12.25 + 8.4 + 7.23) \div 3 = 27.88 \div 3 = 9.293\dots \text{ rounds to } 9.3 \text{ based on the } 2 \text{ significant digits for the } 8.4 \text{ value.}$$

- 8.1.6 **Unit Conversion** – Unit conversions are obtained by multiplying the numerical value by the appropriate conversion factor. In most cases the product of the unconverted numerical value and the factor will be a numerical value with a number of digits that exceeds the number of significant digits of the unconverted numerical value and usually implies an accuracy not warranted by the original value. Conversion of quantities should be handled with careful regard to the implied correspondence between the accuracy of the data and the number of digits. In all conversions, the number of significant digits retained should be such that accuracy is neither sacrificed nor exaggerated.
- 8.1.7 Proper conversion procedure is to multiply the specified numerical value by the conversion factor exactly and then round to the appropriate number of significant digits that is consistent with the maximum possible rounding error of the unconverted numerical value or its accuracy before conversion. In many cases, the conversion factor is not uncertain or is known to a large number of digits. The conversion is treated as in 8.1.5
- 8.1.8 Some conversion factors have extreme levels of precision. In this case, use at least one digit more in conversion factors or constants than is in the measured value with the least number of significant digits.
- 8.1.9 Normally, convert temperatures expressed in a whole number of degrees Fahrenheit or degrees Rankine to the nearest 0.5 K (or degree Celsius). As with other quantities, the

number of significant digits to retain will depend upon the implied accuracy of the original value.

- 8.1.10 The practical aspect of measuring must be considered when using SI equivalents. If a scale having divisions of 1/16 in was suitable for making the original measurements, a metric scale having divisions of 1 mm is suitable for measuring in SI units. Similarly, a gage or caliper graduated in divisions of 0.02 mm is comparable to one graduated in divisions of 0.001 in. Analogous situations exist in the measurement of mass, force, and other quantities.
- 8.1.11 This method depends on first establishing the intended precision or accuracy of the quantity as a necessary guide to the number of digits to retain. This precision should relate to the number of digits in the original, but in many cases this is not a reliable indicator. The number 1.1875 may be the accurate decimalization of 1 3/16, which could have been expressed as 1.19. On the other hand, the number 2 may mean "about 2," or it may mean a very accurate value of 2, which should have been written 2.000.
- 8.1.12 The intended precision of a value must be determined before converting. This estimate of intended precision should never be smaller than the accuracy of measurement, but it should usually be smaller than one-tenth the tolerance, if one exists. After estimating the precision, the converted value should be rounded to a minimum number of significant digits such that a unit of the last place is equal to or smaller than the converted precision.

Examples:

(1) A stirring rod is 6 in long. If the precision of the length of the rod is estimated to be about 1/2 in ($\pm 1/4$ in), the precision is 12.7 mm. The converted value of 152.4 mm should be rounded to the nearest 10 mm, which results in a length of 150 mm.

(2) The test pressure is 200 lbf/in² (psi) $\pm$ 15 lbf/in² (psi). Since one-tenth of the total tolerance is 3 lbf/in² (20.68 kPa), the converted value should be rounded to the nearest 10 kPa. Thus, 1378.9514 kPa $\pm$ 103.421 35 kPa becomes 1380 kPa $\pm$ 100 kPa.

- 8.1.13 **Fraction Conversion** – Fractions consist of an integer divided by another integer. The integers are exact and are treated as though they consist of an infinite number of additional significant digits. The decimal equivalent is then multiplied by the conversion factor, also an exact value. When the converted decimal equivalent is calculated, the number of significant digits should be consistent with the precision of the measurement, i.e. is the 1/4 in. measured to the closest 1/8, 1/16, or 0.001 in., etc.

Fraction, in.	Decimal Equivalent*, in.	Decimal Equivalent, mm
$1/2 = 2/4 = 4/8 = 8/16 = 16/32 = 32/64$	0.500000....	12.7
$1/4 = 2/8 = 4/16 = 8/32 = 16/64$	0.250000...	6.35 rounded to 6.4
$1/8 = 2/16 = 4/32 = 8/64$	0.125000...	3.175 rounded to 3.2
$1/16 = 2/32 = 4/64$	0.0625000...	1.5875 rounded to 1.6
$1/32 = 2/64$	0.0312500....	0.79375 rounded to 0.8
1/64	0.0156250.... rounded to 0.016 (measured to the nearest 0.001 in.)	0.396875 rounded to 0.40

* - Number of significant digits is unknown without knowledge of the precision of the measurement.

- 8.1.14 For converting inches (in.) to millimeters (mm) the general rule is to round to one less decimal place than the original inch dimension. When converting from millimeters to inches, the rule is to round to two more decimal places than the original value.

9.0 ROUNDING:

- 9.1.1 There are several rules available for rounding of numbers. The recommended rule for numbers resulting from measurements and calculations performed on measured numbers is:
- A. When the digit next beyond the last digit to be retained is less than 5, the last digit retained is unchanged. For example: 9.43 rounds to 9.4.
 - B. When the digit next beyond the last digit to be retained is greater than 5, increase the last digit retained by 1. For example: 9.46 rounds to 9.5.
 - C. When the digit next beyond the last digit to be retained is 5, and there are no digits beyond this 5, or only zeroes, increase the last digit retained by 1 if it is odd, leave the digit unchanged if it is even. Increase the last digit retained by 1, if there are non-zero digits beyond this 5. For example: 9.35 rounds to 9.4, 9.65 rounds to 9.6, and 9.651 rounds to 9.7.
- 9.1.2 Never round in the middle of a calculation. Always wait until the final answer to round.
- 9.1.3 Round converted values to the minimum number of significant digits that will maintain the required accuracy. In certain cases, deviation from this practice to make use of convenient or whole numbers may be feasible, in which case use the word "approximate" following the conversion or some other convention to indicate that the value is approximate. For example:

1 7/8 in = 47.625 mm exactly
 = 47.6 mm normal rounding
 = 47.5 mm (approximate) rounded to preferred number
 = 48 mm (approximate) rounded to whole number
 Or 1 7/8 in (47.5 mm) with an accompanying statement such as "Values stated without parentheses are the requirement. Values in parentheses are explanatory or approximate information."

REFERENCES:

For more information on scientific notation:

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